



Matrices : RACE-01

Single Correct Question

1. The number of different orders of a matrix having 12 elements is
(A) 3 (B) 1 (C) 6 (D) None of these

MMT234

2. Construct a 3×2 matrix whose element are given by $a_{ij} = \frac{1}{3} |2i + j|$

(A) $\begin{bmatrix} 1 & 5 \\ 4 & 3 \\ 3 & 2 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 4 \\ 5 & 3 \\ 3 & 2 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 4 \\ 5 & 3 \\ 3 & 2 \end{bmatrix}$ (D) none of these

MMT235

3. Which of the following statement are true:
1. Every identity matrix, I of order n is an upper triangular matrix.
2. Every identity matrix, I of order n is a lower triangular matrix.
(A) Only 1 (B) Only 2 (C) Both 1 and 2 (D) None of these

MMT236

4. Consider the following statements:
1. Every zero matrix is a square matrix.
2. A matrix has a numerical value.
3. A unit matrix is a diagonal matrix.
Which of the above statements is/are correct?

(A) 2 Only (B) 3 Only (C) 2 and 3 (D) 1 and 3

MMT237

5. $\begin{bmatrix} x^2 + x & x \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ -x + 1 & x \end{bmatrix} = \begin{bmatrix} 0 & -2 \\ 5 & 1 \end{bmatrix}$ then x is equal to
(A) -1 (B) 2 (C) 1 (D) No value of x

MMT238

6. If $2A + B = \begin{bmatrix} 6 & 4 \\ 6 & -11 \end{bmatrix}$ and $A - B = \begin{bmatrix} 0 & 2 \\ 6 & 2 \end{bmatrix}$, then $A =$
(A) $\begin{bmatrix} -2 & 2 \\ 4 & -3 \end{bmatrix}$ (B) $\begin{bmatrix} 2 & 0 \\ 4 & -3 \end{bmatrix}$ (C) $\begin{bmatrix} 2 & 2 \\ 4 & -3 \end{bmatrix}$ (D) none of these

MMT239

7. If A and B are square matrices of order 3 such that $|A| = -1$, $|B| = 3$, then $|3AB|$ is equal to -
(A) -9 (B) -81 (C) -27 (D) 81

MMT240

8. If $\begin{bmatrix} 2x+y & 2 & x-2y \\ a-b & 2a+b & -3 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 4 \\ 4 & -1 & -3 \end{bmatrix}$, then find the value of $x + y + a + b$
 (A) 1 (B) -1 (C) -3 (D) 5

MMT241

9. Find the value of x : $2 \begin{bmatrix} 3 & 1 & -2 \\ -1 & -3 & 4 \end{bmatrix} + x \begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 2 & -2 \\ -8 & -14 & -2 \end{bmatrix}$
 (A) -2 (B) 2 (C) 3 (D) -3

MMT242

10. If $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $B = \begin{bmatrix} -5 & 4 & 0 \\ 0 & 2 & -1 \\ 1 & -3 & 2 \end{bmatrix}$, then
 (A) $AB = \begin{bmatrix} -5 & 8 & 0 \\ 0 & 4 & -2 \\ 3 & -9 & 6 \end{bmatrix}$ (B) $AB = \begin{bmatrix} -2 & -1 & 4 \end{bmatrix}$
 (C) $AB = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ (D) AB does not exist

MMT243

11. If $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $j = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ then B equal to
 (A) $I \cos \theta + j \sin \theta$ (B) $I \cos \theta - j \sin \theta$ (C) $I \sin \theta + j \cos \theta$ (D) $-I \cos \theta + j \cos \theta$

MMT244

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	B	C	B	A	C	B	B	A	D
Question	11									
Answer	A									



Matrices : RACE-02

Single Correct Question

1. If $A = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$, where $i = \sqrt{-1}$ and $x \in N$, then A^{4x} equals -

(A) $\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$

MMT245

2. For $\theta = \frac{3\pi}{5}$, let $B = [b_{ij}]$ be a square matrix of 2nd order such that

$$b_{ij} = \begin{cases} \cos \theta & , i = j \\ \cos\left(\frac{j\pi}{2} + \theta\right) & , i > j \\ \sin\left(\frac{j\pi}{2} - \theta\right) & , i < j \end{cases}$$

then trace of matrix B^5 is equal to -

(A) -2 (B) -1 (C) 1 (D) 2

MMT246

3. If A and B are 3×3 matrices such that $AB = A$ and $BA = B$, then-

(A) $A^2 = A, B^2 \neq B$ (B) $A^2 \neq A, B^2 = B$
(C) $A^2 = A, B^2 = B$ (D) $A^2 \neq A, B^2 \neq B$

MMT247

4. If A, B are two square matrices such that $AB = A, BA = B$ then $(A + B)^7$ equals

(A) $A + B$ (B) $2^7(A + B)$ (C) $2^6(A + B)$ (D) $2^8(A + B)$

MMT248

Integer Type Question

5. Let A be a square matrix satisfying $A^2 = 2I$ and $\sum_{k=1}^{2n} A^k = 63A(A + I)$, then the value of n is
(where I denotes identity matrix)

MMT249

6. If $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, find k so that $A^2 = kA - 2I$.

MMT250

7. Let $A = \begin{bmatrix} \alpha^2(a-2\lambda) & 0 & 3 \\ -1 & \alpha(b-3\lambda) & 1 \\ 0 & 5 & (c-4\lambda) \end{bmatrix}$; $B = \begin{bmatrix} \beta^2(a-2\lambda) & -5 & -3 \\ 7 & \beta(b-3\lambda) & 7 \\ 8 & 9 & (c-4\lambda) \end{bmatrix}$ and

$$C = \begin{bmatrix} \gamma^2(a-2\lambda) & -1 & 3 \\ 0 & \gamma(b-3\lambda) & 7 \\ 8 & -3 & (c-4\lambda) \end{bmatrix}, \text{ where } \alpha, \beta, \gamma \text{ are real \& distinct such that}$$

$\text{tr}(A) = \text{tr}(B) = \text{tr}(C) = -1$, where $\lambda \in N$, then minimum possible value of $a + b + c$ is

MMT251

8. Let $A = \begin{bmatrix} -3 & 0 & 2 \\ 1 & x & 5 \\ -2 & 0 & x^2 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ b \\ -1 \end{bmatrix}$ and $C = [3 \ 5 \ 1]$, then the number of integral value(s) of 'b' for which

$\text{Tr}(ABC) \leq -18 \forall x \in R$ is (are)

MMT252

Subjective Type Question

9. If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then using principle of mathematical induction or otherwise prove that

$$A^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix} \text{ where } n \text{ is a positive integer.}$$

MMT253

10. If $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, show that $A^2 = 3A$ and $A^4 = 27A$. Deduce that $A^3 - A = 2A(A + I)$

MMT254

11. (a) $A = \begin{bmatrix} 2 & 3 \\ 5 & -1 \\ 7 & 8 \end{bmatrix}$ then write A^T , (b) $A = \begin{bmatrix} 5 & 1 & 7 \\ 6 & 8 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 5 & 1 \end{bmatrix}$ then find $A + B^T$.

MMT255

ANSWER KEY

Question	1	2	3	4	5	6	7	8	
Answer	C	A	C	C	6	1	8	5	
Question	11								
Answer	(a) $\begin{bmatrix} 2 & 5 & 7 \\ 3 & -1 & 8 \end{bmatrix}$ (b) $\begin{bmatrix} 7 & 2 & 12 \\ 9 & 10 & 10 \end{bmatrix}$								



Matrices : RACE-03

Single Correct Question

1. The matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ -1 & -2 & -3 \end{bmatrix}$ is

- (A) idempotent matrix (B) involutory matrix
(C) nilpotent matrix (D) periodic matrix

MMT256

2. If $A, B, A + I, A + B$ are idempotent matrices (where A and B are square matrices of same order & I is identity matrix of same order), then value of $|AB - BA|$ (where $|C|$ denotes determinant of matrix C) is -
(A) 0 (B) 1 (C) 2 (D) cannot be determined

MMT257

3. If A is nilpotent matrix of index 2, then $A(I_2 + A)^{51}$ is equal to (where I_2 is the identity matrix of order 2) -
(A) A^{51} (B) I_2 (C) Null matrix (D) A

MMT258

4. If $A = \begin{bmatrix} a & b \\ c & 1-a \end{bmatrix}$ is idempotent matrix and $bc = 1/4$ then the value of a is
(A) 1 (B) -1 (C) 1/2 (D) -1/2

MMT259

5. If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$, then $(BB^T A)^5$ is equal to

(Where X^T denotes transpose of matrix X)

- (A) $\begin{bmatrix} 2+\sqrt{3} & 1 \\ -1 & 2-\sqrt{3} \end{bmatrix}$ (B) $\frac{1}{2} \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 5 & 1 \\ 0 & 1 \end{bmatrix}$

MMT260

6. Let B be a square matrix such that $|B| = 1$ and $2A + B = B^T - A^T$, then $|A + B|$ is -
(A) 0 (B) -1 (C) 1 (D) 2

MMT261

Matrices

Multiple Correct Question

7. If A and B are square matrices such that $AB = B$ and $BA = A$, then which of the following is/are always true ?
- (A) A is an idempotent matrix (B) B is an involutory matrix
(C) $A^6 + B^6 = A^8 + B^8$ (D) $A^2 + B^2 = A + I$

MMT262

Integer Type Question

8. Consider the two matrices A and B where $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$; $B = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$. Let $n(A)$ denotes the number of elements in A and $n(XY) = 0$, when the two matrices X and Y are not conformable for multiplication. If $C = (AB)(B'A)$; $D = (B'A)(AB)$ then, find the value of $\left(\frac{n(C)(|D|^2 + n(D))}{n(A) - n(B)} \right)$

MMT263

Subjective Type Question

9. If the matrix A is involutory, show that $\frac{1}{2}(I + A)$ and $\frac{1}{2}(I - A)$ are idempotent and $\frac{1}{2}(I + A) \cdot \frac{1}{2}(I - A) = 0$.

MMT264

10. Show that the matrix $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$ is an orthogonal matrix.

MMT265

11. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying $AA^T = 9I$ where I is 3×3 identity matrix, then the ordered pair (a, b) is equal to

MMT266

ANSWER KEY

Question	1	2	3	4	5	6	7	8	11
Answer	C	A	D	C	C	C	A, C	650	$(-2, -1)$



Matrices : RACE-04

Single Correct Question

1. If A is a matrix of order 3×3 such that $A^2 + A = A^T$, then which of the following is wrong -
 (A) $|A| = 0$ (B) $|A| \neq 0$
 (C) A^2 is a skew symmetric matrix (D) A is singular matrix

MMT267

2. Let A and B be 3×3 matrices of real number where A is symmetric, B is skew symmetric and $(A + B)(A - B) = (A - B)(A + B)$ if $(AB)^T = (-1)^k AB$, then possible value of k is -
 (A) 3 (B) 4 (C) 6 (D) 2

MMT268

3. If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 2 \\ 1 & 1 & 3 \end{bmatrix}$, then $\text{tr}(A^3 \text{adj}(A^2))$, is

(where $\text{tr}(A)$ and $\text{adj}(A)$ denote trace and adjoint of matrix A) -

- (A) 6 (B) 24 (C) 36 (D) 12

MMT269

4. If A is a square matrix of order n with $|A| = 9$ and $|\text{adj}(\text{adj}(\text{adj}A))| = 3^{16}$, then number of elements in A , is -
 (A) 9 (B) 4 (C) 5 (D) 2

MMT270

Multiple Correct Question

5. Let $A = \begin{bmatrix} 0 & \tan \theta & 2 \\ -\tan \theta & 0 & \sec \theta \\ -2 & -\sec \theta & 0 \end{bmatrix}$, where $\theta \neq (2n+1)\frac{\pi}{2}, (n \in I)$ & $B = \sum_{r=1}^5 A^{(2r-1)}$ then which of the

following statement is correct

- (A) A is skew symmetric matrix (B) B is skew symmetric matrix
 (C) $(A + B)$ is skew symmetric matrix (D) $(A + B)$ is symmetric matrix

MMT271

6. Let $A = a_{ij}$ be a matrix of order 3 where $a_{ij} = \begin{cases} x & \text{if } i = j, x \in R \\ 1 & \text{if } |i - j| = 1 \\ 0 & \text{otherwise} \end{cases}$, then which of the following

hold(s) good ?

- (A) for $x = 2$, A is a diagonal matrix.
 (B) A is a symmetric matrix
 (C) for $x = 2$, $\det A$ has the value equal to 6
 (D) Let $f(x) = \det A$, then the function $f(x)$ has both local maxima and local minima.

MMT272

Subjective Type Question

7. If A and B are non-singular matrices of order three such that $\text{adj}(AB) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & p & 1 \\ 1 & 1 & p \end{bmatrix}$ and

$$|B^2 \text{adj}(A)| = p^2 - 3, \text{ then } p \text{ is}$$

MMT273

8. $A_{3 \times 3}$ is a matrix such that $|A| = a$, $B = (\text{adj } A)$ such that $|B| = b$. Find the value of $(ab^2 + a^2b + 1)S$

$$\text{where } \frac{1}{2}S = \frac{a}{b} + \frac{a^2}{b^3} + \frac{a^3}{b^5} + \dots \text{ up to } \infty, \text{ and } a = 3.$$

MMT274

9. If A be a square matrix of the order 5 and $B = \text{Adj}(A)$ then find $\text{Adj}(5A)$.

MMT275

10. If A be a square matrix of order 4 and $|A| = 3$ then find $\text{adj}(\text{adj } A)$.

MMT276

11. Express matrix $A = \begin{bmatrix} 2 & 5 & 7 \\ 9 & -7 & 2 \\ 1 & -1 & 0 \end{bmatrix}$ as a sum of a symmetric and a skew symmetric matrix.

MMT277

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	B	A	A,B,C	B,D	2	225	625B	9A



Matrices : RACE-05

Single Correct Question

1. If 'A' is a square matrix such that $A^2 = I$ then A^{-1} is equal to -
 (A) $A + I$ (B) A (C) 0 (D) $2A$
MMT278
2. Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and $10B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix}$. If B is the inverse of matrix A , then α is-
 (A) -2 (B) -1 (C) 2 (D) 5
MMT279
3. $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$ then let us define a function $f(x) = \det. (A^T A^{-1})$ then which of the following can not be the value of $\underbrace{f(f(f(\dots\dots\dots f(x))))}_{n \text{ times}}$ is ($n \geq 2$)
 (A) $f^n(x)$ (B) 1 (C) $f^{n-1}(x)$ (D) $nf(x)$
MMT280
4. If A is a square matrix such that $A^5 = O$ and $A^n \neq O$ for $n = 1, 2, 3, 4$, then $(I - A)^{-1}$ is equal to (where O is a null matrix)-
 (A) $I + A$ (B) $I + A + A^2$ (C) $I + A^{-1}$ (D) $I + A + A^2 + A^3 + A^4$
MMT281
5. If A and B are non-singular symmetric matrices of the same order such that $AB = BA$ and $P = (AB^{-1})^T$ & $Q = (A^{-1}B)^T$ then $(PQ)^{-1}$ is equal to -
 (A) $AB^{-1}AB$ (B) $A^{-1}B^{-1}AB$ (C) $A^{-1}BA^{-1}B$ (D) A^2B^2
MMT282

Multiple Correct Question

6. Let $A^{-1} = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 2 & 1 \\ 2 & 1 & -1 \end{bmatrix}$, then -
 (A) $7|A| = \frac{1}{2}$ (B) $|\text{adj } A| = \frac{1}{196}$
 (C) $\text{trace}(\text{adj } A) = -\frac{1}{7}$ (D) Matrix A is a symmetric matrix
MMT283

7. If $A = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 2 & 0 \\ 4 & 5 & 3 \end{bmatrix}$, then which of the following is(are) true ?

(trace of A denotes sum of principal diagonal elements of A)

(A) A is invertible

(B) $\text{trace}(\text{adj}(\text{adj}(A))) = 144$

(C) $\text{trace}(\text{adj}(\text{adj}(A))) = 8$

(D) $|\text{adj } A|$ is less than 400

MMT284

Integer Type Question

8. If A is a non-zero idempotent matrix satisfying $\left(I - \frac{2}{5}A\right)^{-1} = I - \alpha A$ where I is the unit matrix of the same order as that of A then the absolute value of 3α is

MMT285

9. Let A, B and $A + B$ are non-singular matrices of order 3×3 satisfying

$$A^{-1} + B^{-1} = (A + B)^{-1} \text{ and } |AB^{-1}| \in R \text{ then value of } \frac{|A|}{|B|} \text{ is}$$

MMT286

10. Let P, Q, R be invertible matrices of second order such that $A = PQ^{-1}$, $B = QR^{-1}$, $C = RP^{-1}$, then $\det(ABC + BCA + CAB)$ is

MMT287

11. An invertible matrix A of order 3 satisfies the relation $A = A^{-1} + 2I$, (where I denotes identity matrix). The value of $|A - I| \cdot |A + I| \cdot |A - 2I|$ is

MMT288

Subjective Type Question

12. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$; $B = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}$; $C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$, then solve the following matrix equation.

(a) $AX = B - I$

(b) $(B - I)X = IC$

(c) $CX = A$

MMT289

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	D	D	D	B	B,C,D	A,B,D	2	1	9
Question	11									
Answer	8									

12. (a) $X = \begin{bmatrix} -3 & -3 \\ 5 & 2 \end{bmatrix}$, (b) $X = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$, (c) no solution



Matrices : RACE-06

Single Correct Question

1. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ is zero divisor of $f(x) = x^2 - 4x - 5$, then A^3 is equal to

(A) $\begin{bmatrix} 41 & 42 & 42 \\ 42 & 41 & 42 \\ 42 & 42 & 41 \end{bmatrix}$ (B) $\begin{bmatrix} 42 & 41 & 42 \\ 41 & 42 & 41 \\ 44 & 41 & 42 \end{bmatrix}$ (C) $\begin{bmatrix} 43 & 42 & 41 \\ 44 & 41 & 42 \\ 42 & 43 & 43 \end{bmatrix}$ (D) $\begin{bmatrix} 41 & 42 & 41 \\ 42 & 41 & 42 \\ 42 & 42 & 41 \end{bmatrix}$

MMT290

2. Let A be a 3×3 matrix such that $a_{11} = a_{33} = 2$ & all the other $a_{ij} = 1$. Let $A^{-1} = xA^2 + yA + zI$, then the value of $(x + y + z)$ where I is a unit matrix of order 3, is -
(A) 0 (B) 1 (C) 2 (D) 3

MMT291

3. The system of equations $x + 2y - 3z = 1$, $x - y + 4z = 0$, $2x + y + z = 1$ has -
(A) only two solutions (B) only one solution
(C) no solution (D) infinitely many solutions

MMT292

4. The system of equations $x + y + z = 8$, $x - y + 2z = 6$, $3x + 5y - 7z = 14$ has -
(A) Unique solution (B) infinite number of solutions
(C) no solution (D) none of these

MMT293

5. Consider the system of equations $x - 2y + 3z = -1$, $x - 3y + 4z = 1$ and $x + y + 2z = k$ has exactly three solutions, then -
(A) $k = 3$ (B) $k = 0$ (C) $k = -9$ (D) $k = \phi$

MMT294

Integer Type Question

6. If the system of linear equations $x + 2ay + az = 0$, $x + 3by + bz = 0$, $x + 4cy + cz = 0$, has a non-trivial solution & $a, b, c > 0$ then the minimum value of $\left(\frac{a^3 + c^3}{b^3}\right)$ is

MMT295

7. Square matrices B of second order are formed using some or all the possible values of k , when system of linear equations $kx + 4y + z = 0$, $4x + ky + 2z = 0$ & $2x + 2y + z = 0$, has infinite solution, then number of singular matrices B is

MMT296

Subjective Type Question

8. Determine the characteristic roots of the matrix A . Hence find the trace and determinant value of A .

Where $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ and also prove that $A^3 - 18A^2 + 45A = \mathbf{0}$.

MMT297

9. Find the characteristic polynomial equation of $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \\ -1 & 0 & 3 \end{bmatrix}$

MMT298

10. If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then show that the matrix A is a root of the polynomial $f(x) = x^3 - 6x^2 + 7x + 2$.

MMT299

ANSWER KEY

Question	1	2	3	4	5	6	7	
Answer	A	B	D	A	D	2	6	
Question	8				9			
Answer	$\lambda = 0,3$ and $15 \operatorname{tr}(A) = 18, A = 0$				$x^3 - 6x^2 + 13x - 11 = 0$			